

AI Project Foundation

Iris Flower Classification Using Machine Learning

Multi-Class Supervised Classification System

Domain: Artificial Intelligence & Machine Learning

Project Type: Multi-Class Classification

Dataset: Iris Flower Dataset | 150 Samples | 4 Features | 3 Classes

Technology Stack: Python · Scikit-Learn · Pandas · Matplotlib · Jupyter · GitHub

Target Accuracy: > 95%

Timeline: 6-Day Sprint Plan

Prepared by the Iris AI Project Team

1. Project Selection

The Iris Flower Classification project is a foundational machine learning initiative designed to introduce core AI concepts through a well-studied, real-world dataset. By leveraging the famous Iris dataset — first introduced by statistician Ronald Fisher in 1936 — this project demonstrates the end-to-end machine learning pipeline from raw data ingestion to accurate species prediction.

Project Title: Iris Flower Classification Using Machine Learning

Project Domain: Artificial Intelligence & Machine Learning

Project Type: Multi-Class Supervised Classification

Algorithms Explored: K-Nearest Neighbors (KNN), Random Forest, Decision Tree, SVM

Target Metric: Classification Accuracy > 95%

The Iris dataset is considered the "Hello World" of machine learning — compact, clean, and ideal for benchmarking classification algorithms.

2. Problem Statement

Flower species identification has traditionally depended on trained botanists and manual observation — a process that is time-consuming, subjective, and difficult to scale. With thousands of plant species distributed globally, automated identification systems offer a compelling solution.

This project addresses the classification problem: given four measurable physical attributes of an Iris flower (sepal length, sepal width, petal length, and petal width), the system must correctly identify the species as one of three classes:

- Iris Setosa — distinguishable by notably smaller petal dimensions
- Iris Versicolor — intermediate measurements across all four features
- Iris Virginica — characterized by larger petals and sepals

Why This Problem Matters

Automated plant classification has practical relevance across multiple domains:

- Agricultural monitoring: enabling precision farming through species tracking
- Botanical research: accelerating large-scale biodiversity surveys
- Conservation: detecting invasive species in ecosystems
- Education: providing accessible, data-driven botany tools

Challenge

The primary challenge is that Iris Versicolor and Iris Virginica share overlapping feature ranges, making them harder to separate than Iris Setosa. A robust model must learn to distinguish these subtle differences using data-driven boundaries.

3. AI Use Case Identification

AI Application Type: Supervised Machine Learning – Multi-Class Classification

Business Use Cases

- Automated plant species identification in mobile botanical apps
- Agricultural monitoring systems for crop quality control
- Botanical research assistance and taxonomy automation
- Biodiversity analysis and ecosystem health reporting
- Intelligent classification systems embedded in IoT sensors
- Educational platforms for interactive AI/ML demonstrations

Input / Output Specification

Role	Variable	Description
Input	Sepal Length (cm)	Length of the sepal in centimetres
Input	Sepal Width (cm)	Width of the sepal in centimetres
Input	Petal Length (cm)	Length of the petal in centimetres
Input	Petal Width (cm)	Width of the petal in centimetres
Output	Species Label	Iris Setosa / Versicolor / Virginica

4. Objectives & Expected Outcomes

Project Objectives

- Understand the end-to-end machine learning workflow and best practices
- Perform exploratory data analysis (EDA) to uncover patterns in the Iris dataset
- Build and train multiple supervised classification models
- Compare algorithm performance using standardised evaluation metrics
- Achieve species prediction accuracy exceeding 95%
- Evaluate model robustness using cross-validation and confusion matrices
- Document all findings, code, and results for reproducibility

Expected Outcomes

Deliverable	Description
Trained Classification Model	Best-performing algorithm saved as iris_model.pkl
Model Accuracy	Greater than 95% on test split and cross-validation
EDA Report	Visualisations: pair plots, heatmaps, distribution charts
Comparison Report	Side-by-side metrics for KNN, Random Forest, Decision Tree, SVM
GitHub Repository	Complete codebase with notebooks, scripts, dataset, and README
Final Presentation	6-slide summary deck for stakeholder review
Project Documentation	Full technical write-up of methods, results, and conclusions

5. Dataset Identification

Dataset Overview

Attribute	Detail
Dataset Name	Iris Flower Dataset
Source	UCI Machine Learning Repository / Scikit-Learn
Total Samples	150
Samples per Class	50 each (perfectly balanced)
Number of Features	4 numerical features
Target Classes	3 (Setosa, Versicolor, Virginica)
Missing Values	None
Data Type	Continuous numerical values
File Format	CSV (IRIS.csv)

Feature Descriptions

Feature Name	Type	Unit	Relevance
SepalLengthCm	Continuous	Centimetres	Moderate discriminator
SepalWidthCm	Continuous	Centimetres	Weaker discriminator
PetalLengthCm	Continuous	Centimetres	Strong discriminator
PetalWidthCm	Continuous	Centimetres	Strongest discriminator

Petal dimensions are the most discriminating features, particularly for separating Iris Setosa from the other two species. Sepal width shows the highest variance and most overlap between Versicolor and Virginica.

6. GitHub Repository Setup

Repository Name: iris-flower-classification-ml

Visibility: Public

License: MIT License

Repository Directory Structure

Path	Purpose
dataset/IRIS.csv	Raw Iris dataset in CSV format
notebooks/iris_analysis.ipynb	Jupyter Notebook: EDA and model training walkthrough
src/preprocessing.py	Data loading, cleaning, and feature scaling scripts
src/train_model.py	Model training, hyperparameter tuning, and saving
src/prediction.py	Load saved model and predict species for new inputs
models/iris_model.pkl	Serialised best-performing trained model (Pickle)
images/	EDA plots, confusion matrices, and performance charts
README.md	Project overview, setup instructions, and usage guide
requirements.txt	Python dependency list for reproducible environment
presentation/	Final slide deck and project summary report

README.md Key Sections

- Project description and motivation
- Installation guide: `pip install -r requirements.txt`
- Usage instructions for running notebooks and scripts
- Model performance summary and benchmark results
- Contributor information and license details

7. Project Architecture Design

ML Pipeline Overview

The project follows a structured, sequential machine learning pipeline with clear stage gates at each phase:

Step	Stage	Key Activities
01	Data Ingestion	Load IRIS.csv using Pandas; inspect shape, dtypes, and head
02	Data Preprocessing	Validate missing values; encode labels; apply StandardScaler
03	Exploratory Data Analysis	Pair plots, correlation heatmap, box plots, class distribution
04	Feature Analysis	Identify discriminating features; assess multicollinearity
05	Train-Test Split	80/20 stratified split ensuring balanced class representation
06	Model Training	Train KNN, Random Forest, Decision Tree, SVM classifiers
07	Hyperparameter Tuning	GridSearchCV / cross-validation for optimal parameters
08	Model Evaluation	Accuracy, precision, recall, F1-score, confusion matrix
09	Model Serialisation	Save best model with Joblib (iris_model.pkl)
10	Species Prediction	Load model and classify new flower measurements

Technology Stack

Category	Technology	Usage
Language	Python 3.x	Core development language
Data Manipulation	Pandas, NumPy	DataFrame handling and numerical computation
Visualisation	Matplotlib, Seaborn	EDA plots and performance charts
Machine Learning	Scikit-Learn	Model training, evaluation, and tuning
Development IDE	Jupyter Notebook	Interactive exploration and documentation
Version Control	GitHub	Code hosting, versioning, and collaboration
Model Persistence	Joblib / Pickle	Save and load trained models

8. Project Plan & Timeline

The project is structured as a 6-day sprint, with each day dedicated to a specific milestone in the machine learning lifecycle:

Day	Phase	Tasks & Deliverables
Day 1	Setup & Dataset Analysis	Configure Python environment · Clone GitHub repo · Load IRIS.csv · Inspect data shape, null values, and class distribution · Generate initial data summary report
Day 2	Data Preprocessing & EDA	Apply StandardScaler normalisation · Encode target labels · Visualise feature distributions · Create pair plots, correlation heatmap, and box plots · Document EDA findings
Day 3	Feature Analysis & Visualisation	Assess feature importance using correlation scores · Visualise feature separability per class · Build scatter plots for sepal vs petal dimensions · Identify top discriminating features
Day 4	Model Training	Implement 80/20 stratified train-test split · Train KNN, Random Forest, Decision Tree, and SVM · Apply GridSearchCV for hyperparameter optimisation · Save best model as iris_model.pkl
Day 5	Model Evaluation	Generate classification reports (precision, recall, F1) · Plot confusion matrices for each algorithm · Run k-fold cross-validation · Compare all models and select champion model
Day 6	Documentation & Presentation	Write README.md · Finalise Jupyter Notebook with inline documentation · Create 6-slide presentation deck · Push all code and assets to GitHub · Prepare final project report

9. Presentation Summary

The Iris Flower Classification project demonstrates the complete supervised machine learning lifecycle applied to a classic botanical dataset. The following points summarise the key takeaways for a stakeholder presentation:

Core Project Overview

- Objective: classify Iris flowers into Setosa, Versicolor, or Virginica using four physical measurements
- Dataset: 150 samples, 4 features, 3 balanced classes, zero missing values
- Approach: supervised multi-class classification with multiple algorithm comparison

Algorithms Evaluated

Algorithm	Expected Accuracy	Strengths	Weaknesses
K-Nearest Neighbors (KNN)	~97%	Simple, no training phase	Slow on large data

Random Forest	~96–98%	Robust, feature importance	Less interpretable
Decision Tree	~93–96%	Highly interpretable	Prone to overfitting
Support Vector Machine	~97%	Effective in high dimensions	Kernel tuning needed

Key Insights

- Petal length and petal width are the most powerful predictors of Iris species
- Iris Setosa is linearly separable from the other two species using petal features alone
- Iris Versicolor and Virginica require non-linear boundaries for accurate separation
- All four algorithms are expected to exceed the 95% accuracy target on this dataset

Success Criteria

- Model accuracy > 95% on held-out test set
- Cross-validation F1-score > 0.95 across all three classes
- Complete, documented code published on GitHub
- Reproducible Jupyter Notebook from data loading to final prediction

This project establishes a reusable ML template that can be adapted for more complex classification tasks — including multi-feature crop classification, medical diagnosis, and industrial quality control.